Sana Niroomand

Tehran, Iran | sana.niroomand84@sharif.edu | +98 990 171 72 32 | Sana Niroomand | SanaNiroomand

Education

BS **Sharif University of Technology**, Computer Engineering — Sept 2023 – Present

- GPA: 3.78/4.0(18.88/20.0)
- **Notable Coursework:** Intelligent Analysis of Biomedical Images using Deep Learning, Computational Drug Design, Machine Learning, Introduction to Bioinformatics, AI, Linear Algebra, Probability and Statistics, Calculus, Discrete Structures, Algorithms Design and Data Structures, Advanced Programming, Object Orientation

HS **Farzanegan SAMPAD School**, Mathematics & Physics — Sept 2017 – July 2023

- GPA: 3.93/4.0(19.51/20.0)
- **SAMPAD (NODET)** National Organization for Development of Exceptional Talents

Projects

Interpretable and Uncertainty-aware Multiple Instance Learning under Weak Supervision

Currently developing a Multiple Instance Learning (MIL) framework for medical image analysis by adapting the architecture proposed in the 2025 MICCAI paper Multi-scale Attention-based Multiple Instance Learning for Breast Cancer Diagnosis to additional CT/MRI datasets. Implementing multi-scale attention-based instance aggregation under weak supervision, with a focus on model interpretability and uncertainty estimation. Building a flexible training and evaluation pipeline to support PI-CAI dataset for prostate cancer predicition, due to the fact that breast and prostate cancers share overlapping genetic drivers and signaling pathways, despite arising in different organs.

Yara - Intelligent Medical Assistant (Ongoing)

Yara is an AI-native assistant that creates a smart online record for every patient. It listens to doctor–patient conversations, automatically generates structured clinical notes and summaries into the patient history, and keeps all past visits, labs, and medications in one place. Through an intelligent, context aware chat, doctors can ask Yara personalized questions—such as what to prescribe or what the patient’s full history shows—based on that patient’s record.

Transcriptomic Analysis of Psoriasis and Psoriatic Arthritis Skin

Analytical bioinformatics project aimed at reproducing the results of the study “Differences in transcriptional changes in psoriasis and psoriatic arthritis skin with immunoglobulin gene enrichment in psoriatic arthritis”. The project focuses on replicating differential gene expression and analyses using public RNA-seq datasets.

Semantic Road Segmentation using Deep Learning

Developed semantic segmentation models in Python/PyTorch for road detection on the Massachusetts Roads Dataset. Implemented U-Net, Attention U-Net, and Residual Attention U-Net from scratch, with data loading, preprocessing, and augmentation for satellite TIFF imagery. Built a full training pipeline with BCE-Dice loss, Adam optimizer, learning-rate scheduling, early stopping, and model checkpointing. Evaluated performance using Accuracy, IoU, and Dice Score, and visualized predictions vs. ground-truth masks.

Gwent: The Witcher Card Game

Developed an implementation of Gwent using Object-Oriented Programming principles, JavaFX for graphics, and basic networking features. (Project completed as part of the Advanced Programming course in collaboration with Arash Ghavami and Parsa Norouzi Manesh.)

Atomic Bomber

Created a clone of the Atomic Bomber game to practice computer graphics concepts using JavaFX.

Neogit: A Simple Git Implementation

Designed and built a lightweight version control system inspired by Git, implemented in C.

MIPS Processor Implementation with I/O

Developed a single-cycle MIPS processor in Logisim-evolution with a custom UART-inspired serial communication protocol.

Optimized Matrix Computation on AMD64 Architecture
Implemented matrix multiplication and inner product calculators using both SISD (Single Instruction Single Data) and SIMD (Single Instruction Multiple Data) instructions.
Designed and developed two different algorithmic approaches for each calculation to compare performance and efficiency.

Preparation of a Botanical Antimicrobial Spray Using Plant-Derived Compounds
Designed and tested a natural antimicrobial spray using plant extracts, validated through microbial culture assays to ensure efficacy and reduced side effects compared to chemical sprays.
Developed in collaboration with Nastaran Nasrian and Fatemeh Goudarzi.
Achievements:

- Awarded the **Gold Medal** at an international exhibition in South Korea
- Ranked **2nd Place** in the Innovations and Inventions Track at the Avicenna (Ibn Sina) Festival
- Achieved **Provincial Distinguished Rank** in the 5th Khwarizmi Youth Festival

Presentation of a Proposal for a Mobile Application Aimed at Enhancing Neighborly Relations
The application offered building management features and sought to revitalize the diminishing interpersonal connections among neighbors in Iran.
Developed in collaboration with Hasti Cheraghchi.

- **Achievement:** Provincial Distinguished Rank in the Khwarizmi Festival

Honors and Awards

- **5th rank** out of around 190 students studying Computer Engineering from 2023 at Sharif University, February 2024
- **7th rank** out of 188 students studying Computer Engineering from 2023 at Sharif University, August 2024
- **197th national rank** out of 147,317 students (among the top 0.14%) in the Mathematics field in the Iranian Nationwide University Entrance Exam for Bachelor Studies (Concours), Iran 2023
- **Provincial Distinguished Rank** in the 5th Khwarizmi Youth Festival, Tehran 2019 (in collaboration with Nastaran Nasrian)
- **Provincial Distinguished Rank** in the Khwarizmi Festival, Tehran 2021 (in collaboration with Hasti Cheraghchi)
- **2nd Place** in the Inventions & Achievements Track at the Avicenna (Ibn Sina) Festival, Iran 2018 (in collaboration with Nastaran Nasrian and Fatemeh Goudarzi)

Research Experiences

Deep Learning Applications in Agriculture — Vision-Language Models for Plant Disease Detection — February 2025 – May 2025

- Researcher at the Deep Learning Lab, focused on advancing agricultural technologies using deep learning techniques
  Supervisors: Prof. Rohban, Prof. Sabzi
- Exploring the integration of vision-language models (VLMs) for real-time plant disease diagnosis based on image prompts

Volunteering

- **Member of education deputy at RSG-Iran** An initiative to provide a platform for Iranian students to train, network, and endeavor in the field of computational biology and bioinformatics in Iran. Under the supervision of Dr. Ali Sharifi-Zarchi.
- **Advanced Programming** TA for OOP and Design Patterns assignments (Supervisor: Prof. Fazli)